

MATH 2211, SPRING 2026: MIDTERM 2

This exam has 6 (six) printed pages. Please show all your work. Outside materials are NOT permitted for use on this test. Do not spend more than 10 minutes on any one part of a problem. If you get stuck for more than a few minutes, just move on to the next part. *You should feel free to use the results of earlier problems in your solutions to later ones, regardless of whether you have finished them or not.* You have 50 minutes to work on the test. Write your answers in the space provided.

Name _____

	Score
Problem 1	/10
Problem 2	/10
Problem 3	/10
Problem 4	/10
Problem 5	/10
Total	/50

Name _____

- (1) (10 points) **True or false. Write one or two lines justifying your answer:**

- (a) There is an invertible matrix $A \in M_{n \times n}(F)$ with linearly dependent columns.

False. The columns of an invertible matrix form a basis and are in particular linearly independent

- (b) There exists a linear transformation $T : \text{Poly}_{\leq 5}(F) \rightarrow F^7$ with $\dim \text{im } T = 4$ and $\dim \ker T = 3$.

False. By rank-nullity, we have
 $\dim \text{im } T + \dim \ker T = \dim \text{Poly}_{\leq 5}(F) = 6$

$4+3=7 \neq 6$

- (c) If a matrix $A \in M_{n \times n}(F)$ is invertible, then so is A^T .

True $\det(A^T) = \det(A)$ by transpose invariance
 $A \text{ invertible} \Leftrightarrow \det(A) \neq 0 \Leftrightarrow \det(A^T) \neq 0 \Leftrightarrow A^T \text{ invertible}$

- (d) Every linear transformation $T : F^2 \rightarrow F^2$ is diagonalizable.

False T given by the matrix $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$
 is a counterexample

- (e) If $A \in M_{n \times n}(F)$ is invertible then $\det(A^{-1}) = \det(A)^{-1}$.

True $\det(A^{-1}A) = \det(I_n) = 1$
 $\parallel$
 $\det(A^{-1}) \cdot \det(A)$
 $\Rightarrow \det(A^{-1}) = \det(A)^{-1}$

Name _____

(2) (10 points) Consider the linear transformation

$$T: \mathbb{Q}^3 \rightarrow \mathbb{Q}^3$$

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} \mapsto \begin{bmatrix} a - c \\ b + c \\ a + b \end{bmatrix}$$

Find bases for the kernel and image of T .

T is given by matrix

$$[T(e_1) \ T(e_2) \ T(e_3)]$$

$$= \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} \xrightarrow{R_3 - R_1} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$\xrightarrow{R_3 - R_2}$$

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

ker T : solutions to
 $a - c = 0$
 $b + c = 0$

$$= \left\{ \begin{bmatrix} t \\ -t \\ t \end{bmatrix} = t \cdot \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} : t \in \mathbb{Q} \right\} \quad \text{Basis} = \left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$$

im T : Basis is given by columns where
 RREF has leading 1s

$$\leadsto \text{Basis} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$$

Name _____

- (3) (10 points) Suppose that $(v_1, \dots, v_n)$ is a basis for F^n , and that $A \in M_{n \times n}(F)$. Show that the following are equivalent:

(a) $\det(A) \neq 0$

(b) $(Av_1, \dots, Av_n)$ is also a basis for F^n .

$(v_1, \dots, v_n)$ basis
 $\Leftrightarrow B = [v_1 \ v_2 \ \dots \ v_n]$
 is invertible

$\det(A) \neq 0 \Leftrightarrow A$ is invertible
 $\Leftrightarrow AB$ is invertible
 $\Leftrightarrow A[v_1 \ v_2 \ \dots \ v_n]$ is invertible
 $\Leftrightarrow [Av_1 \ Av_2 \ \dots \ Av_n]$ is invertible
 $\Leftrightarrow (Av_1, Av_2, \dots, Av_n)$ is a basis

To see this equivalence, note that $A = (AB)B^{-1}$

Name _____

- (4) (10 points) Consider the matrix $C = \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix} \in M_{2 \times 2}(\mathbb{Q})$ with associated linear transformation $T: \mathbb{Q}^2 \rightarrow \mathbb{Q}^2$.

(a) Compute the characteristic polynomial and eigenvalues of C .

$$\begin{aligned} \chi_C(x) &= x^2 - \text{Tr}(C)x + \det(C) \\ &= x^2 - 5x + 4 = (x-4)(x-1) \end{aligned}$$

$$\text{e. values} = 4, 1$$

- (b) Show that T is diagonalizable, and use this to compute C^{10} .

Compute eigenspaces E_4, E_1

$$\begin{aligned} E_4 &= \ker(C - 4I_2) = \ker\left(\begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix}\right) \\ &= \left\{ t \begin{bmatrix} 1 \\ 1 \end{bmatrix} : t \in \mathbb{Q} \right\} \end{aligned}$$

$$\begin{aligned} E_1 &= \ker(C - I_2) = \ker\left(\begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}\right) \\ &= \left\{ t \begin{bmatrix} 1 \\ -2 \end{bmatrix} : t \in \mathbb{Q} \right\} \end{aligned}$$

Eigenbasis:

$$\begin{aligned} v_1 &= \begin{bmatrix} 1 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix} \\ B &= (v_1, v_2) \\ [T]_B &= \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix} \\ P &= \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}: \text{change of basis} \end{aligned}$$

$$\begin{aligned} C &= P[T]_B P^{-1} = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}^{-1} \\ &= \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2/3 & 1/3 \\ 1/3 & -1/3 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} C^{10} &= \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 4^{10} & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2/3 & 1/3 \\ 1/3 & -1/3 \end{bmatrix} = \begin{bmatrix} 4^{10} & 1 \\ 4^{10} & -2 \end{bmatrix} \begin{bmatrix} 2/3 & 1/3 \\ 1/3 & -1/3 \end{bmatrix} \\ &= \frac{1}{3} \begin{bmatrix} 2 \cdot 4^{10} + 1 & 4^{10} - 1 \\ 2 \cdot 4^{10} - 2 & 4^{10} - 2 \end{bmatrix} \end{aligned}$$

Name _____

(5) (10 points) Let a, b, c be three *distinct* elements of a field F . Show that

$$\det \begin{pmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{pmatrix} \neq 0$$

This is equivalent to showing that the matrix is **invertible**

Can do this using **row reduction**

$$\begin{bmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{bmatrix} \xrightarrow[\substack{R_2 - R_1 \\ R_3 - R_1}]{} \begin{bmatrix} 1 & a & a^2 \\ 0 & b-a & b^2-a^2 \\ 0 & c-a & c^2-a^2 \end{bmatrix}$$

permissible because $c-a \neq 0$ $b-a \neq 0$

$$(c-a)^{-1} \cdot R_3 \quad \downarrow \quad (b-a)^{-1} \cdot R_2$$

$$\begin{bmatrix} 1 & a & a^2 \\ 0 & 1 & b+a \\ 0 & 0 & c-b \end{bmatrix} \xleftarrow{R_3 - R_2} \begin{bmatrix} 1 & a & a^2 \\ 0 & 1 & b+a \\ 0 & 1 & c+a \end{bmatrix}$$

OK since $c-b \neq 0$

$$(c-b)^{-1} \cdot R_3$$

$$\begin{bmatrix} 1 & a & a^2 \\ 0 & 1 & b+a \\ 0 & 0 & 1 \end{bmatrix}$$

Invertible!!

Req: This actually shows

$$\det = (b-a)(c-a)(c-b)$$